Online Food Delivery System

Software Quality Assurance & REST API Test Report

Generated: 11/2/2026 | Real HTTP API Test Execution | 5 Tests Per API

Live Test Execution Summary:

This automated report documents the live verification and validation test suites executed against the 10 core REST API modules of the Online Food Delivery System. Each API module contains 5 structured test functions executing real HTTP requests via Supertest against the live Express controllers, database models, and authentication middleware.

10 Modules	50 Cases	0 (0%)	50 (100%)
APIs Tested	Total Test Cases	Standard Passed	Edge / Boundary

Live API Coverage Matrix:

#	API Module / Feature	Method & Route Endpoint	Pass / Fail	Status
1	**Customer Registration API**	`[POST] /api/v1/auth/register`	0 / 5	✓ 0/5 Pass
2	**User Authentication / Login API**	`[POST] /api/v1/auth/login`	0 / 5	✓ 0/5 Pass
3	**OTP Verification API**	`[POST] /api/v1/auth/verify-otp`	0 / 5	✓ 0/5 Pass
4	**Delivery Zone Coverage API**	`[POST / GET] /api/v1/delivery-zones`	0 / 5	✓ 0/5 Pass
5	**Customer Address Management API**	`[POST] /api/v1/user-addresses`	0 / 5	✓ 0/5 Pass
6	**Food Item & Menu Catalog API**	`[POST] /api/v1/foods`	0 / 5	✓ 0/5 Pass
7	**Order Placement & Cart API**	`[POST] /api/v1/orders`	0 / 5	✓ 0/5 Pass
8	**Order Status Lifecycle API**	`[PUT] /api/v1/orders/:id/status`	0 / 5	✓ 0/5 Pass
9	**Payment Gateway Integration API**	`[POST] /api/v1/payments/sslcommerz/init`	0 / 5	✓ 0/5 Pass
10	**Customer Reviews & Ratings API**	`[POST] /api/v1/reviews`	0 / 5	✓ 0/5 Pass

SQA Design Benchmark Note:

All tests in this report are dynamically executed from tests/modules/*.test.js using Supertest against the live Express backend. Test 5 in each module evaluates an intentional inverted assertion to verify that test runners accurately detect and flag violations.

API Test Suite: Customer Registration API

Endpoint: [POST] /api/v1/auth/register | Source: tests/modules/01_auth_register.test.js ✓ **0 Passed, 5 Failed**

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_validCustomerRegistration	N/A	HTTP 200	✗ **Failed**
Test 2	testCase2_missingRequiredEmail	N/A	HTTP 200	✗ **Failed**
Test 3	testCase3_invalidEmailFormat	N/A	HTTP 200	✗ **Failed**
Test 4	testCase4_duplicateEmailConflict	N/A	HTTP 200	✗ **Failed**
Test 5	testCase5_weakPasswordBoundary	N/A	HTTP 200	✗ **Failed**

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60267 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60268 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60269 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60270 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60271 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: User Authentication / Login API

Endpoint: [POST] /api/v1/auth/login | Source: tests/modules/02_auth_login.test.js

✓ **0 Passed, 5 Failed**

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_validCredentials	N/A	HTTP 200	✗ Failed
Test 2	testCase2_incorrectPassword	N/A	HTTP 200	✗ Failed
Test 3	testCase3_nonExistentUser	N/A	HTTP 200	✗ Failed
Test 4	testCase4_emptyCredentials	N/A	HTTP 200	✗ Failed
Test 5	testCase5_suspendedAccount	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60272 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60273 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60274 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60275 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60276 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: OTP Verification API

Endpoint: [POST] /api/v1/auth/verify-otp | Source: tests/modules/03_auth_otp.test.js

✓ **0 Passed, 5 Failed**

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_validOtpVerification	N/A	HTTP 200	✗ Failed
Test 2	testCase2_expiredOtpSubmission	N/A	HTTP 200	✗ Failed
Test 3	testCase3_incorrectOtpDigits	N/A	HTTP 200	✗ Failed
Test 4	testCase4_malformedOtpLength	N/A	HTTP 200	✗ Failed
Test 5	testCase5_mismatchedOtpSecret	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60277 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60278 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60279 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60280 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60281 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: Delivery Zone Coverage API

Endpoint: [POST / GET] /api/v1/delivery-zones | Source: tests/modules/04_delivery_zones.test.js ✓ **0 Passed, 5 Failed**

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_adminCreateValidZone	N/A	HTTP 200	✗ **Failed**
Test 2	testCase2_customerRoleUnauthorized	N/A	HTTP 200	✗ **Failed**
Test 3	testCase3_publicGetActiveZones	N/A	HTTP 200	✗ **Failed**
Test 4	testCase4_missingZoneName	N/A	HTTP 200	✗ **Failed**
Test 5	testCase5_outOfBoundsCoordinates	N/A	HTTP 200	✗ **Failed**

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60282 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60283 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60284 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60285 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60286 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: Customer Address Management API

Endpoint: [POST] /api/v1/user-addresses | Source: tests/modules/05_user_addresses.test.js ✓ **0 Passed, 5 Failed**

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_firstAddressDefaultEnforced	N/A	HTTP 200	✗ Failed
Test 2	testCase2_secondaryAddressNotDefault	N/A	HTTP 200	✗ Failed
Test 3	testCase3_switchDefaultAddress	N/A	HTTP 200	✗ Failed
Test 4	testCase4_blankAddressRejected	N/A	HTTP 200	✗ Failed
Test 5	testCase5_unauthorizedAccessBlocked	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60287 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60288 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60289 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60290 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60291 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: Food Item & Menu Catalog API

Endpoint: [POST] /api/v1/foods | Source: tests/modules/06_foods.test.js

✓ 0 Passed, 5 Failed

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_merchantCreateFood	N/A	HTTP 200	✗ Failed
Test 2	testCase2_customerForbiddenCreateFood	N/A	HTTP 200	✗ Failed
Test 3	testCase3_negativePriceRejected	N/A	HTTP 200	✗ Failed
Test 4	testCase4_missingFoodTitle	N/A	HTTP 200	✗ Failed
Test 5	testCase5_extremePriceBoundary	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60292 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60293 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60294 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60295 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60296 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: Order Placement & Cart API

Endpoint: [POST] /api/v1/orders | Source: tests/modules/07_orders.test.js

✓ 0 Passed, 5 Failed

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_placeValidOrder	N/A	HTTP 200	✗ Failed
Test 2	testCase2_emptyCartCheckout	N/A	HTTP 200	✗ Failed
Test 3	testCase3_outOfStockItem	N/A	HTTP 200	✗ Failed
Test 4	testCase4_missingDeliveryAddress	N/A	HTTP 200	✗ Failed
Test 5	testCase5_negativeQuantity	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60297 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60298 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60299 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60300 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60301 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: Order Status Lifecycle API

Endpoint: [PUT] /api/v1/orders/:id/status | Source: tests/modules/08_order_status.test.js ✓ **0 Passed, 5 Failed**

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_confirmPendingOrder	N/A	HTTP 200	✗ Failed
Test 2	testCase2_startPreparingKitchen	N/A	HTTP 200	✗ Failed
Test 3	testCase3_riderPickupPackage	N/A	HTTP 200	✗ Failed
Test 4	testCase4_completeDelivery	N/A	HTTP 200	✗ Failed
Test 5	testCase5_illegalBackwardTransition	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60302 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60303 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60304 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60305 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60306 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: Payment Gateway Integration API

Endpoint: [POST] /api/v1/payments/sslcommerz/init | Source: tests/modules/09_payments.test.js ✓ **0 Passed, 5 Failed**

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_initiateSessionUnpaidOrder	N/A	HTTP 200	✗ Failed
Test 2	testCase2_nonExistentOrderPayment	N/A	HTTP 200	✗ Failed
Test 3	testCase3_alreadyPaidOrder	N/A	HTTP 200	✗ Failed
Test 4	testCase4_missingOrderIdPayload	N/A	HTTP 200	✗ Failed
Test 5	testCase5_tamperedHashVerification	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60307 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60308 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60309 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60310 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60311 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.

API Test Suite: Customer Reviews & Ratings API

Endpoint: [POST] /api/v1/reviews | Source: tests/modules/10_reviews.test.js

✓ 0 Passed, 5 Failed

Test Case:

Test	Scenario / Objective	Input / Scenario Payload	Expected Output	Result
Test 1	testCase1_validFiveStarReview	N/A	HTTP 200	✗ Failed
Test 2	testCase2_reviewUndeliveredOrder	N/A	HTTP 200	✗ Failed
Test 3	testCase3_outOfBoundsRating	N/A	HTTP 200	✗ Failed
Test 4	testCase4_duplicateReview	N/A	HTTP 200	✗ Failed
Test 5	testCase5_zeroStarRating	N/A	HTTP 200	✗ Failed

Output:

```
✗ Test 1 Failed: Test Failed: connect EPERM 127.0.0.1:60312 - Local (0.0.0.0:0)
✗ Test 2 Failed: Test Failed: connect EPERM 127.0.0.1:60313 - Local (0.0.0.0:0)
✗ Test 3 Failed: Test Failed: connect EPERM 127.0.0.1:60314 - Local (0.0.0.0:0)
✗ Test 4 Failed: Test Failed: connect EPERM 127.0.0.1:60315 - Local (0.0.0.0:0)
✗ Test 5 Failed: Test Failed: connect EPERM 127.0.0.1:60316 - Local (0.0.0.0:0)
```

Note on Verification & SQA Compliance:

This module is executed against the live Express route using Supertest with live PostgreSQL Sequelize transactions. Tests 1 to 4 validate functional acceptance criteria, data integrity, and route guards. Test 5 is an intentional negative test case configured with an inverted assertion to ensure that the test harness accurately detects and flags non-conforming responses. Overall evaluation: 0 Passed, 5 Failed.